

Separable Differential Equations with Initial Conditions

Separating variables, finding the constant from an initial condition, and using the particular solution

Directions

- Separate the variables first: get every y (with dy) on one side and every x (with dx) on the other before integrating.
- One constant of integration is enough. Put $+C$ on the x side only.
- Substitute the initial values as soon as you have integrated, then solve for C before rearranging.
- Where a decimal is asked for, round only at the very last step.

1. Separate the variables in $\frac{dy}{dx} = 5x^4y$ and integrate both sides. Write your result in the form $\ln |y| = F(x) + C$.

Answer: _____

2. Separate the variables in $\frac{dy}{dx} = \frac{4x}{y}$ and integrate both sides. Write your result in the form $\frac{1}{2}y^2 = G(x) + C$.

Answer: _____

3. Solve the initial value problem $\frac{dy}{dx} = 2xy$, $y(0) = 5$. Write the particular solution $y = f(x)$, then evaluate $y(1)$ to three decimal places.

Answer: _____

4. A quantity y satisfies $\frac{dy}{dx} = \frac{y}{4}$ with $y(0) = 12$ and $y > 0$. Solve the initial value problem, then find the value of x at which $y = 48$. Give an exact value and a decimal to three places.

Answer: _____

- 5.** Separate the variables in $\frac{dy}{dx} = \frac{\cos x}{2y}$ and integrate both sides to obtain $y^2 = \sin x + C$.
- (a) Carry out the separation and the integration.
 - (b) Explain why this result does not determine a single function $y = f(x)$ until an initial condition is given.

Answer: _____

- 6.** Solve the initial value problem $\frac{dy}{dx} = \frac{3x^2}{y}$, $y(1) = 4$, for the branch with $y > 0$. Write the particular solution $y = f(x)$, then evaluate $y(2)$ to three decimal places.

Answer: _____

7. A colony of bacteria has mass P (in thousands of cells) after t days and satisfies $\frac{dP}{dt} = 0.08P$ with $P(0) = 25$. Solve the initial value problem and find $P(10)$ to two decimal places.

Answer: _____

8. Solve the initial value problem $\frac{dy}{dx} = e^{x-y}$, $y(0) = \ln 3$. (Hint: $e^{x-y} = e^x e^{-y}$.) Write the particular solution $y = f(x)$, then evaluate $y(\ln 5)$ to three decimal places.

Answer: _____

9. Water drains from a tank so that the depth h , measured in centimetres, satisfies $\frac{dh}{dt} = -\frac{3}{2\sqrt{h}}$, where t is in minutes. The depth at $t = 0$ is 36 centimetres. Find the number of minutes until the tank is empty ($h = 0$), to the nearest tenth of a minute.

Answer: _____

10. Consider the initial value problem $\frac{dy}{dx} = y^2 \sin x$, $y(0) = \frac{1}{2}$.
- (a) Find the particular solution $y = f(x)$, and confirm that it gives $y(0) = \frac{1}{2}$.
 - (b) Explain why the largest interval containing $x = 0$ on which this solution is defined stops before $x = \pi$.

Answer: _____